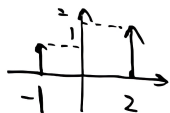


1-7(4) 1-20 1-22 1-28 2-2

1-7(4)

$$f(t) [\delta(t+1) + \delta(t-2)] = f(t) \delta(t+1) + f(t) \delta(t-2)$$



1-20

$$(1) \int_{-4}^4 [\delta(t) + \delta(t-1)] dt$$

$$= 2 + 1 = 3$$

(2)

$$\int_{-\pi}^{\pi} (t \cos t) \delta(t - \frac{\pi}{2}) dt$$

$$= \int_{-\pi}^{\pi} \delta(t - \frac{\pi}{2}) dt$$

$$= 1$$

(3)

$$\int_{-2\pi}^{2\pi} (1+t) \delta(\cos t) dt$$

$$= \int_{-2\pi}^{2\pi} \left[(1+t) \delta(t - \frac{\pi}{2}) + (1+t) \delta(t - \frac{3\pi}{2}) + (1+t) \delta(t + \frac{\pi}{2}) + (1+t) \delta(t + \frac{3\pi}{2}) \right] dt$$

$$= 1 + \frac{\pi}{2} + 1 - \frac{\pi}{2} + 1 + \frac{3\pi}{2} + 1 - \frac{3\pi}{2} = 4$$

1-22

$$x_2(t) = u(t-1) - u(t-2)$$

$$y_2(t) = y(t-1) - y(t-2)$$

$$= e^{-(t-1)} u(t-1) + u(t-1) - [e^{-(t-2)} u(t-2) + u(t-2)]$$

1-28

$$x_2(t) = \frac{dx}{dt}$$

$$y_2(t) = \frac{dy}{dt} = \delta(t) - ae^{-at} u(t)$$

2-2

(1)

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} k u[k] \delta[n-2-k]$$

$$= \sum_{k=0}^{+\infty} k \delta[n-2-k]$$

$$= (n-2) u[n-2]$$

(2)

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} 2^k u[k] u[n-k]$$

$$= \sum_{k=0}^n 2^k \quad (n \geq 0)$$

$$= \frac{1-2^{n+1}}{1-2} (2^{n+1} - 1) u[n]$$

(3)

$$x[n] * h[n] = \sum_{k=0}^n 2^k \quad n \geq 0$$

$$= \frac{1}{2} (2^{n+1} - 1) u[n]$$

(3)

$$x[n] = 2^n u[n-1]$$

$$h[n] = \frac{1}{2^n} u[n-1]$$

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} 2^k u[-k-1] \frac{1}{2^{n-k}} u[n-k-1]$$

$$= \begin{cases} \sum_{k=-\infty}^{n-1} 2^{2k-n} & n < 0 \\ \sum_{k=-\infty}^{-1} 2^{2k-n} & n \geq 0 \end{cases}$$

$$= \frac{1}{3} 2^{-n}$$

(4)

$$x[n] = a^n u[n] \quad h[n] = \beta^n u[n] \quad a \neq \beta$$

$$x[n] * h[n]$$

$$= \begin{cases} 0 & n < 0 \\ \sum_{k=0}^n \beta^k \left(\frac{a}{\beta}\right)^k & n \geq 0 \end{cases}$$

$$= \frac{\beta^{n+1} - a^{n+1}}{\beta - a} u[n]$$

(5)

$$x[n] = (-1)^n (u[n] - u[n-8])$$

$$h[n] = u[n] - u[n-8]$$

$$x[n] * h[n] = \begin{cases} \sum_{k=n-7}^0 (-1)^k & 0 \leq n \leq 7 \\ \sum_{k=-7}^n (-1)^k & -7 \leq n < 0 \\ 0 & \text{其它} \end{cases}$$

$$= \begin{cases} \frac{1 - (-1)^{n+1}}{2} & 0 \leq n \leq 7 \\ \frac{-1 + (-1)^{n+1}}{2} & -7 \leq n < 0 \\ 0 & \text{其它} \end{cases}$$

编程题1

1. 请编程实现一维卷积功能，即自己编写MATLAB中的CONV函数，自己写出一个CONV_NEW函数。要求：（1）课堂上采用列表法编写了CONV_NEW，请试着采用卷积公式法，编写CONV_NEW。（2）请用例子检查你写的CONV_NEW是否与CONV的输出是否一致。（3）请用两个10000维数组检测对比你写的CONV_NEW和CONV的时间复杂度。请查资料回答，为什么MATLAB自带的CONV更快一些？

(1)

conv_New.m

```
function C = conv_New(A,B)
% 第一问的解答
C = zeros(1,length(A) + length(B)-1);
for i = 1:length(A)
    C = C + [zeros(1,i-1),A(i)*B, zeros(1,length(A)-i)];
end
```

(2)

```
%检验手写的卷积函数与matlab内置函数%
A = [1,2,3];
B = [4,5];
C1 = conv(A,B);
C2 = conv_New(A,B);
disp('matlab内置函数的卷积结果: ');
disp(C1);
disp('手写函数的卷积结果: ');
disp(C2);
% 结果如下:
% matlab内置函数的卷积结果:
%      4      13      22      15

% 手写函数的卷积结果:
%      4      13      22      15
%      两种卷积计算结果相同
```

(3)

```
A = rand(1,10000);
B = rand(1,10000);
tic
C1 = conv_New(A,B);
toc
tic
C2 = conv(A,B);
```

```
toc
```

```
% MATLAB自带的CONV函数使用了快速傅里叶变换（FFT）算法来计算卷积，  
%这种算法的时间复杂度为 $O(n \log n)$ ，  
%而我写的CONV_NEW函数使用了直接计算卷积的方法，时间复杂度为 $O(n^2)$ 。  
%因此，MATLAB自带的CONV函数在处理大规模数据时更快一些。  
%结果如下%  
% >> question3  
% 历时 0.116491 秒。  
% 历时 0.006995 秒。
```